From Single-Task to Multi-Task Learning for Dual-Barrier CNS Prediction: A Decision-Aware Evaluation

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Abstract

Predicting brain exposure from a blood–brain barrier (BBB) permeability label alone omits active transporter recognition, including P-glycoprotein (P-gp) status. We compared independent graph models with a shared-encoder multi-task graph neural network (GNN) using the project's BBB and P-gp benchmarks. The datasets overlap for 216 molecules, with weak label association (Cramér's $V = 0.064$; Fisher odds ratio 1.42, 95% CI 0.76–2.67). In five-fold out-of-fold evaluation, single-task models achieved ROC-AUCs of 0.9379 (BBB) and 0.9011 (P-gp), compared with 0.9363 and 0.9056 for the multi-task model; these results do not demonstrate a reliable transfer benefit. On the dual-labeled molecules, quadrant exact match was 57.9% for composed single-task models and 56.0% for the multi-task model. A BBB-only rule sent 42 of 45 true BBB^+/P-gp-positive cases to the favorable quadrant. Repository XGBoost screening, feature attribution and docking analyses provide additional context, but are not independent clinical or experimental validation. The P-gp endpoint's substrate interpretation also requires source-level confirmation.

Keywords: blood–brain barrier; P-glycoprotein; multi-task learning; graph neural network; CNS drug discovery; decision-aware evaluation

1 Introduction

The blood–brain barrier (BBB) regulates molecular access from blood to the central nervous system through tightly connected endothelial cells and selective transport processes. Predicting whether a compound is likely to cross this barrier is an established computational task, and graph neural networks (GNNs) have been applied to represent molecules directly as graphs of atoms and bonds. The GCN-BBB study supplied with this project provides the starting point for the initial graph-model replication [1]. Curated datasets such as B3DB provide a larger collection of BBB permeability labels for model development [2].

Permeability does not by itself determine brain exposure. P-glycoprotein (P-gp/ABCB1), expressed at the BBB, can transport recognized substrates back toward the circulation [3]. Loperamide illustrates the relevance of transporter-mediated restriction: P-gp inhibition can

increase its delivery to the brain [4]. Passive permeation and active efflux are distinct processes; neither a BBB^+ call nor a P-gp classifier output alone establishes unbound brain concentration, therapeutic exposure or clinical efficacy.

The project therefore frames CNS prediction as a two-axis classification problem. Each molecule is assigned a BBB label and a P-gp label; their combinations form four quadrants. For this paper, consistent with the authors' clarification, $P\text{-gp}^+$ denotes a molecule annotated as a substrate that is pumped back toward the blood, while $P\text{-gp}^-$ denotes the non-substrate/efflux-negative label. BBB^+/P-gp-negative is a useful screening hypothesis, not a sufficient condition for CNS drug suitability. It does not capture unbound brain exposure, other transporters, metabolism or target engagement.

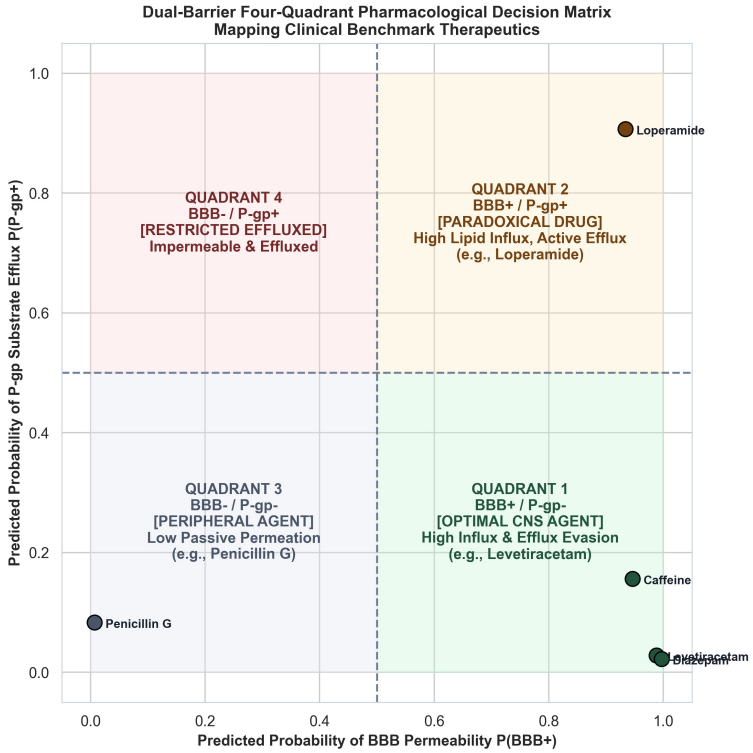


Figure 1: Four-quadrant representation combining BBB permeability and the project's P-gp pipeline labels.

1.1 Study motivation and questions

A BBB model can perform well on its own label while remaining unable to distinguish a permeant P-gp substrate from a permeant non-substrate. The two benchmark labels overlap for 216 molecules, allowing a direct, though limited, examination of this issue. Prior work has also described combined BBB/P-gp classification, so the present study does not claim to introduce the dual-barrier concept [5].

We ask: (i) how much association is observed between the two labels in the shared set; (ii) whether hard parameter sharing improves per-task or joint decision performance over separate models; and (iii) what is lost when a single-axis BBB decision is used in place of a joint classification. We additionally report explainability, approved-drug screening and docking outputs from the repository as supporting analyses, with their scope and limitations stated.

P-gp is an ATP-dependent transporter with two transmembrane domains and a broad ligand-binding cavity. Structural and pharmacological studies describe recognition as multi-specific rather than governed by a single universal motif; molecular size, hydrophobic interactions and other ligand features can contribute in context-dependent ways [6, 7]. This helps explain why passive membrane partitioning cannot be treated as a surrogate for transporter recognition. The present classifiers nevertheless learn benchmark labels, not transporter kinetics or net brain exposure. The project-specific interpretation of the positive class is stated above; the cited Broccatelli publication itself is titled as an inhibitor-prediction study, so exact source-label lineage remains a reproducibility caveat.

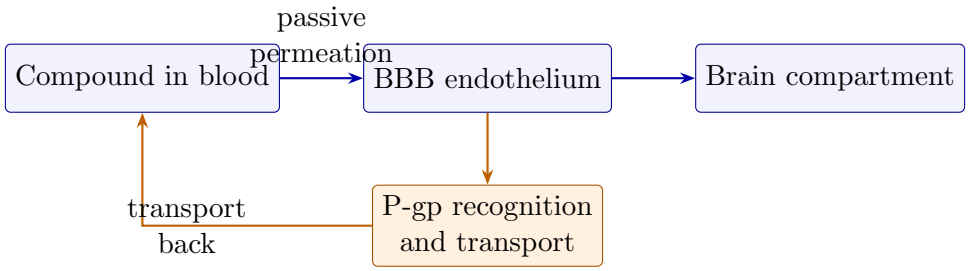


Figure 2: Conceptual distinction between passive permeation and transporter-mediated return.

1.2 Related computational approaches

Early BBB classifiers used molecular descriptors and statistical learning to distinguish compounds reported to penetrate the brain from compounds that did not [8–10]. Fingerprints and physicochemical descriptors remain practical baselines because they encode local substructures and global properties while allowing efficient training on curated data [11, 12]. B3DB subsequently assembled a diverse collection of BBB measurements and labels, creating a widely used resource for prediction and benchmarking [2]. These binary labels are useful for ranking, but they simplify a continuum of permeability and often combine experimental protocols and thresholds.

Graph neural networks offer a complementary representation: message passing updates atom states using neighboring atom and bond attributes, then pools atom states to represent the molecule. The supplied GCN-BBB paper motivated the initial project replication [1]; the present graph arm uses an edge-aware GINE-style network rather than claiming an exact reimplementation of every architectural or data choice in that study. General graph-learning work provides the methodological context for this representation [13, 14].

Transporter-aware prediction is not new. Earlier permeability models explicitly distinguished potential CNS drugs from P-gp substrates [5], and structural studies have since refined understanding of the transporter cavity and its substrate/inhibitor discrimination [7]. The project contribution is therefore not the invention of a dual-barrier concept. Its focused question is empirical: does a shared graph encoder improve the two tasks in this particular, sparsely co-labeled benchmark, and do task-level scores translate into reliable joint quadrant decisions?

1.3 Study objectives and hypotheses

The analysis has four objectives. First, quantify the observed relationship between the two labels among molecules measured on both endpoints, without treating a nonsignificant association

test as proof of independence. Second, compare separate graph models with a hard-parameter-sharing model under the same task definitions and report the limited power of repeated-seed comparisons. Third, evaluate compound-level decision quality on the complete dual-labeled out-of-fold subset, emphasizing errors that advance a BBB-positive efflux substrate as if it were an unimpeded CNS candidate. Fourth, place those results in the context of the repository's independently developed XGBoost screening system, feature ablations, model explanations and exploratory docking.

The motivating transfer hypothesis was that the smaller P-gp task might benefit from a representation learned on the much larger BBB dataset. The counter-hypothesis is that if the measured labels encode weakly associated properties, shared parameters may offer little transferable signal or may create negative transfer. The work treats both as testable expectations, not as established biological explanations.

2 Data Sources and Preparation

The graph-learning benchmark combined the B3DB classification data with the project's P-gp table, distributed through the Therapeutics Data Commons (TDC) ecosystem [2, 15, 16]. Canonical molecular identity was used to merge records, and each task retained an explicit label mask. The graph corpus contained 7,805 BBB-labeled molecules and 1,212 P-gp-labeled molecules, with 216 molecules measured on both tasks, yielding 8,801 unique structures. The validated descriptor/XGBoost branch uses its own processed files (7,805 BBB and 1,205 P-gp structures); those counts differ from the GNN assembly and are kept separate rather than silently combined.

For this paper, positive P-gp labels are interpreted as substrate/efflux-positive and negative labels as non-substrate/efflux-negative, following the project authors' clarification. This is the intended meaning of the repository's task label; the scientific source attribution should still be checked against the exact TDC version and label mapping because the cited Broccatelli article describes inhibitor prediction rather than substrate identification. Results should be read as performance against the project-defined annotation, not as a new experimental measurement of transport.

Input structures were parsed and canonicalized; invalid or unlabelled records were excluded, and duplicate structures with conflicting labels were not resolved by row order. The dual-task assembly represented absent labels with masks rather than imputing them as negative. The graph corpus recorded 1,058 available LogBB values, but continuous LogBB regression was outside the reported experiments. This avoids conflating a thresholded BBB classification target with a continuous permeability estimate.

Dataset partitions were molecule-level: a structure was assigned as a whole and never split into different task-specific train/test identities. Random partitions assess interpolation across a largely shared chemical space; Bemis–Murcko scaffold partitions test transfer to frameworks absent from training [17]. For the GNN single-run comparison, the common random protocol allocated 6,160/1,320/1,321 molecules to train/validation/test, with 33 dual-labeled test cases; the scaffold protocol allocated the same totals, with 48 dual-labeled test cases and no scaffold overlap between train and test. The five-fold random out-of-fold (OOF) analysis was used for joint decision metrics because it supplied held-out predictions for all 216 shared molecules.

Table 1: Graph-learning corpus composition and the dual-labeled contingency table.

Source/stratum	BBB labels	P-gp labels	Unique molecules
B3DB benchmark	7,805	–	7,805
P-gp benchmark	–	1,212	1,212
Shared structures	216	216	216
Outer-joined corpus	7,805	1,212	8,801

Dual-labeled group	P-gp$^+$	P-gp$^-$	Total
BBB$^+$	45	95	140
BBB$^-$	19	57	76
Total	64	152	216

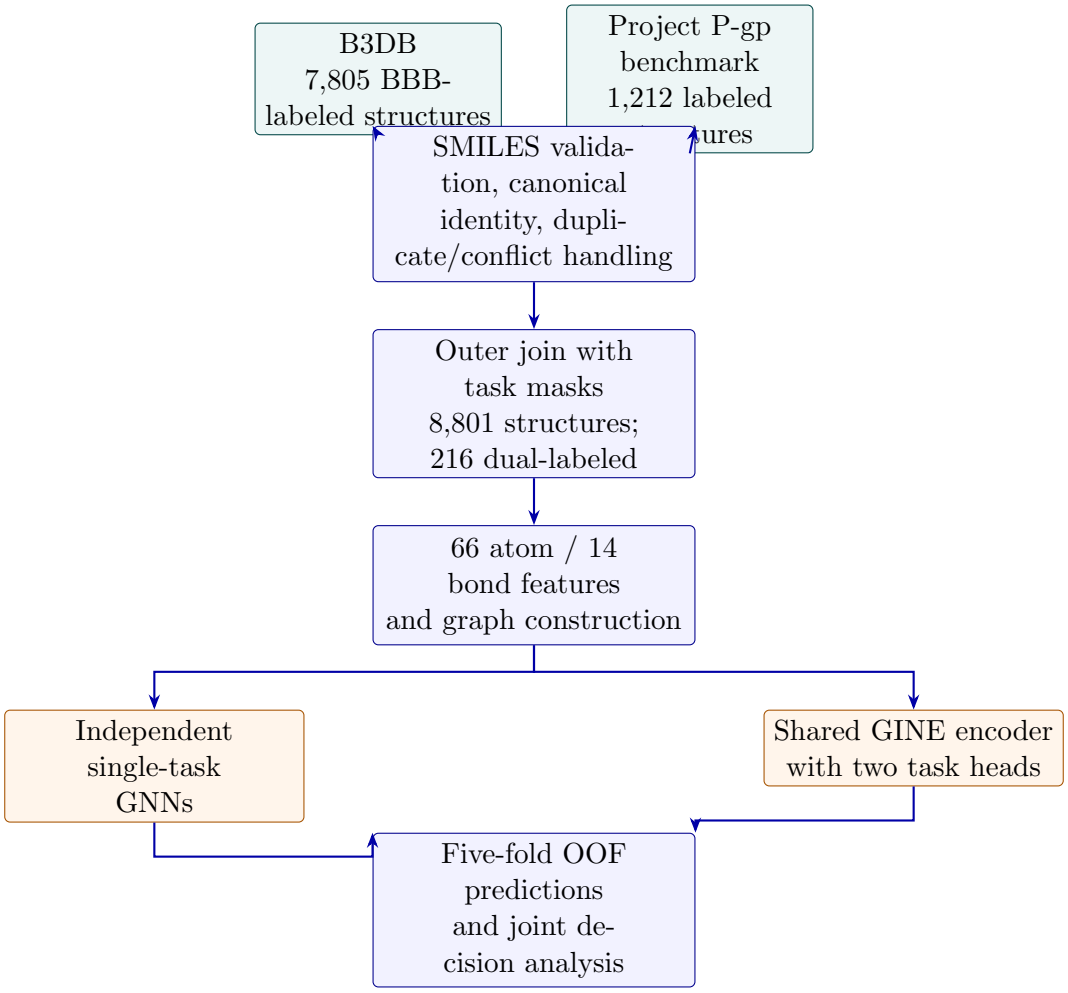


Figure 3: Data assembly and evaluation flow. Every arrow terminates at a box border; task-specific labels remain masked when unobserved.

2.1 Sources of heterogeneity and scope of labels

BBB measurements can reflect different experimental conditions, classification thresholds and compound provenance. B3DB is a curated resource rather than a single uniform permeability experiment; a binary class necessarily discards continuous variation and protocol metadata [2]. P-gp annotations also depend on assay type and curation. A substrate assay, an inhibition assay, a cellular accumulation assay and an in vivo brain-to-plasma measurement are related but not interchangeable endpoints. The project-level label mapping is used consistently in this analysis, but external validation against a second, explicitly documented substrate dataset remains necessary.

The outer join contains 7,589 BBB-only structures and 996 P-gp-only structures in addition to the 216 shared structures. Thus, approximately 97.6% of corpus records have only one observed endpoint. This imbalance is not a missing-at-random guarantee: the two benchmarks were curated separately, and chemical coverage can differ. The masked loss prevents an absent label from becoming a false negative, but it cannot remove dataset-specific selection bias or assay heterogeneity.

3 Model Selection: Development Journey

Model selection followed a sequence of increasingly joint formulations, rather than assuming that the most complex architecture would be best.

Starting point: graph-based BBB prediction. The project began by reproducing a GCN-style BBB classifier based on the supplied GCN-BBB paper and the repository's legacy dataset/code. This established a graph-learning baseline and informed the subsequent expansion from one endpoint to two [1].

Independent endpoint models. The project then developed separate BBB and P-gp classifiers. The screening workflow uses 2,048-bit Morgan fingerprints with six 2D descriptors for BBB, and adds six 3D conformer descriptors for the full P-gp feature set, with XGBoost classifiers producing separate endpoint scores that can be composed into a four-quadrant result [18]. These models provide an independent-task comparison and a practical screening system.

Shared-encoder hypothesis. Because only 216 molecules have both labels, the two tasks were assembled into a molecule-level corpus with a shared graph encoder and task-specific output heads. A masked loss lets each head learn from all molecules labeled for its endpoint without treating a missing label as a negative. This hard-parameter-sharing multi-task GNN adds a second prediction head to a graph representation [19].

Selection outcome. The multi-task model was retained as an experimental comparison, not selected as a superior replacement. Its per-task differences from single-task GNNs were small and inconsistent across random and scaffold splits; increasing the P-gp task weight did not yield a stable improvement. The journey thus motivates reporting both the simpler independent models and the multi-task result, including its null finding.

3.1 Descriptor-based screening models

The repository's deployable screening system uses two independently trained XGBoost classifiers [18]. Both begin with a 2,048-bit Morgan fingerprint (radius 2, conventionally referred to as ECFP4) and six 2D descriptors: molecular weight, calculated LogP, topological polar surface

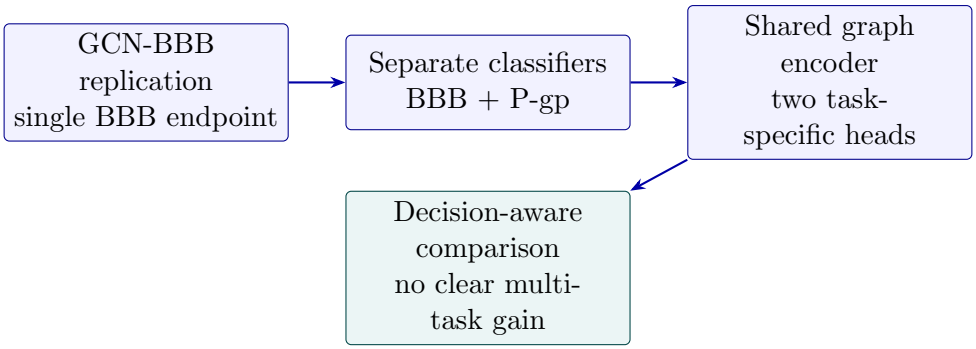


Figure 4: Model-selection path from the single-task baseline to the tested multi-task architecture.

area (TPSA), hydrogen-bond donors, hydrogen-bond acceptors and rotatable bonds. The BBB vector therefore contains 2,054 model features. The P-gp vector adds six geometry descriptors from a generated 3D conformer—radius of gyration, asphericity, eccentricity, spherocity index and the two normalized principal moments of inertia—for 2,060 features. Conformers are generated deterministically using ETKDGv3 with a fixed random seed, then minimized using the project pipeline's force-field workflow [20–22].

The BBB XGBoost model was tuned using Optuna within a train/validation/test design; the final model uses early stopping and evaluates a held-out test partition. The separate P-gp XGBoost model uses the processed P-gp descriptor set and its own training protocol. The two models are not assumed to form a statistically calibrated joint probability. Their probability scores are combined only to construct predicted binary quadrants at the selected threshold, and their validation populations differ from the graph OOF comparison. For that reason, XGBoost values are reported as a separate project arm rather than a controlled architecture contest.

3.2 Graph featurization

For graph learning, each molecule is represented as an undirected chemical graph encoded as directed edges for message passing. The encoded feature mode used in the reported experiments represents each atom by 66 values: one-hot blocks for element, degree, formal charge, hydrogen count, hybridization, chirality and minimum ring size, followed by binary and bounded continuous flags. Each bond has 14 values describing bond type, stereo configuration and binary flags. The graph encoder uses these categorical and local chemical attributes directly; it does not receive the XGBoost Morgan fingerprint vector.

The earlier reference implementation used a legacy 9-atom/7-bond encoding. Experiments in this paper use the newer 66/14 encoded mode, and the parameter count and feature dimensions are recorded with the run reports. This distinction matters for replication: the baseline is inspired by the supplied GCN-BBB reference, but it is not a claim of exact feature parity with its legacy graph input.

3.3 Single-task and multi-task graph architectures

The graph networks use three edge-aware GINE message-passing layers with 256 hidden channels. Each layer applies an MLP-based neighborhood update, batch normalization, ReLU and dropout (0.30); a global mean-pooling operation maps the atom representations to one 256-dimensional graph embedding. A task-specific readout maps this vector through a 32-unit

hidden layer and a two-class output. The single-task models have 364,608 trainable parameters; the two-head shared model has 372,898, a 2.3% increase over the single-head network. The second head branches from the pooled graph embedding, so the two tasks share message passing but retain distinct output layers.

For molecule i, let $m_{it} \in \{0, 1\}$ indicate whether task t has an observed label, and let ℓ_{it} be its class-weighted cross-entropy. The optimized multi-task objective is

$$\mathcal{L}(\theta) = \sum_{t \in \{\text{BBB,Pgp}\}} \alpha_t \frac{\sum_i m_{it} \ell_{it}(\theta)}{\sum_i m_{it}}, \tag{1}$$

with default task weights $\alpha_t = 1$. Class weights are computed from the training partition using inverse class frequency. The denominator is task-specific, so the larger BBB set does not dominate solely by contributing more labeled rows. Missing labels are never converted into negative labels. The shared trunk is therefore exposed to each task's labeled subset, while only observed target values contribute gradients to the corresponding head.

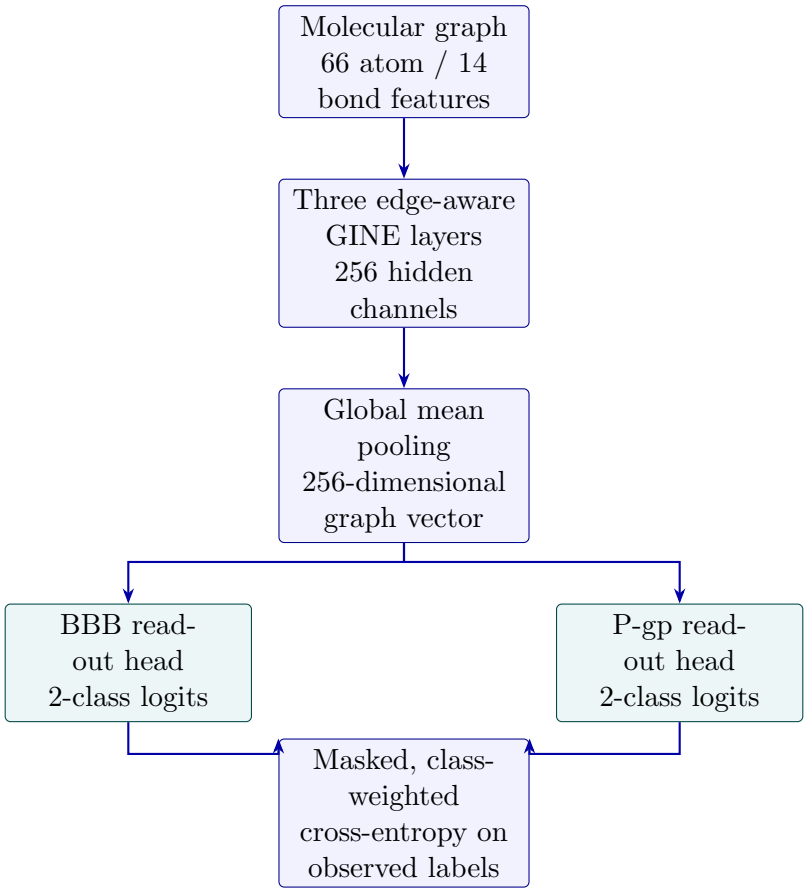


Figure 5: Shared-encoder multi-task GNN. The arrows route from the graph input through the shared trunk into task-specific heads and the masked objective; lines terminate at node borders.

3.4 Optimization and model selection

Training used AdamW with learning rate 5×10^{-4} and weight decay 5×10^{-4}, batch size 64, at most 200 epochs, and early stopping after 20 validation epochs without improvement. A ReduceLROnPlateau scheduler halves the learning rate after eight stagnant epochs, with a floor

of 10^{-6}; gradient norm is clipped at 2.0. The validation selection score is the mean F1 across active tasks. The best validation checkpoint is restored before test evaluation. These values are fixed in the shared training loop so single-task and multi-task experiments use the same implementation.

Table 2: Core graph-network training configuration recorded in the experiment reports.

Setting	Value
Graph encoder	3-layer edge-aware GINE
Hidden width / readout width	256 / 32
Dropout	0.30
Optimizer	AdamW
Learning rate / weight decay	5×10^{-4} / 5×10^{-4}
Batch size / maximum epochs	64 / 200
Early-stopping patience	20 epochs
Validation score	Mean F1 over active task heads
Task weights (baseline MTL)	BBB 1.0; P-gp 1.0
Gradient clipping	Maximum norm 2.0

4 Model Evaluation

Per-task ranking was evaluated using ROC-AUC and area under the precision–recall curve (PR-AUC); thresholded performance was summarized using accuracy, balanced accuracy, precision, recall, specificity, F1 and Matthews correlation coefficient (MCC). ROC-AUC measures ranking over thresholds, while PR-AUC is sensitive to class prevalence. Accuracy alone is insufficient under class imbalance, so threshold-dependent metrics are provided alongside both curves.

4.1 Out-of-fold protocol and uncertainty

For pooled graph-model metrics, every labeled structure received one prediction from a model that was not trained on that structure. Five-fold evaluation covered all 7,805 BBB labels and all 1,212 P-gp labels and produced predictions for all 216 dual-labeled molecules. Within each fold, the training partition was further divided to select the checkpoint using the mean validation F1; predictions were generated on the held-out fold only. Integrity checks confirmed unique OOF assignment and no train/test prediction leakage. The pooled OOF results were bootstrapped 1,000 times for 95% confidence intervals; the intervals quantify uncertainty in the held-out molecule sample and do not capture all sources of model or dataset uncertainty.

The separate random/scaffold XGBoost validation uses its own split-generation and fitting procedure. The scaffold run trains a fresh model and is not a nested, retuned replica of the random-split Optuna configuration. Consequently, random-to-scaffold differences describe the saved project evaluations, not a perfectly controlled split-only experiment. The GNN single-split scaffold values are another distinct evaluation and must not be numerically compared as though they used the XGBoost partitions.

4.2 Joint endpoint and decision metrics

The dual-label contingency table was tested using Pearson's chi-squared statistic, Fisher's exact odds ratio and Cramér's V. The odds-ratio interval was calculated on the 2×2 table. A non-

significant p-value is not interpreted as proof that the endpoints are independent. For quadrant evaluation, the binary predictions on the two axes were combined into Q1 (BBB^+/P-gp^-), Q2 (BBB^+/P-gp^+), Q3 (BBB^-/P-gp^-) and Q4 (BBB^-/P-gp^+). Exact match requires both axis predictions to be correct; Cohen's κ summarizes chance-corrected four-class agreement.

Paradox recall is the fraction of observed Q2 compounds predicted as Q2. Paradox leakage is the fraction of true Q2 compounds predicted as Q1, the progression-eligible quadrant; leakage is therefore a specific consequential error, not the complement of recall. The BBB-only baseline supplies no P-gp prediction and assigns the negative P-gp class by assumption. Confidence intervals for quadrant and paradox metrics were obtained from 1,000 bootstrap resamples of the 216 molecule-level records.

To make the implicit progression decision explicit, we also report a simple cost-sensitive regret:

$$R(\lambda) = \frac{\lambda N_{\text{false progress}} + N_{\text{missed Q1}}}{N}, \tag{2}$$

where λ is the relative cost of advancing a non-Q1 molecule versus missing a true Q1 molecule. This is a transparent sensitivity analysis, not a clinical utility estimate; the repository contains no elicited clinical cost ratio. Reporting a curve over λ avoids presenting one arbitrary cost as universal.

In five-fold out-of-fold evaluation, the single-task GNN reached BBB ROC-AUC 0.9379 (95% CI 0.9323–0.9433) and P-gp ROC-AUC 0.9011 (0.8838–0.9190). The multi-task GNN reached 0.9363 (0.9312–0.9416) and 0.9056 (0.8885–0.9226), respectively. The intervals overlap substantially. For the 216 dual-labeled compounds, decision-aware evaluation additionally measured quadrant exact match, Cohen's κ, paradox recall and the leakage rate into the favorable BBB^+/P-gp non-substrate quadrant.

The following confusion matrices are from the XGBoost models' random held-out test sets; they are not the GNN out-of-fold predictions summarized above.

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Figure 6: XGBoost random held-out test-set confusion matrices for BBB (left) and P-gp (right).

Table 3: Pooled five-fold OOF performance for independent and multi-task GNNs. ROC-AUC confidence intervals use 1,000 bootstrap resamples; other values are point estimates.

Endpoint	Model	n	ROC-AUC (95% CI)	PR-AUC	Bal. Acc.	Specificity	MCC
BBB	Single-task	7,805	0.9379 (0.9323–0.9433)	0.9571	0.8594	0.8192	0.7197
BBB	Multi-task	7,805	0.9363 (0.9312–0.9416)	0.9587	0.8523	0.8077	0.7065
P-gp	Single-task	1,212	0.9011 (0.8838–0.9190)	0.9076	0.8493	0.8407	0.6984
P-gp	Multi-task	1,212	0.9056 (0.8885–0.9226)	0.9181	0.8337	0.8018	0.6698

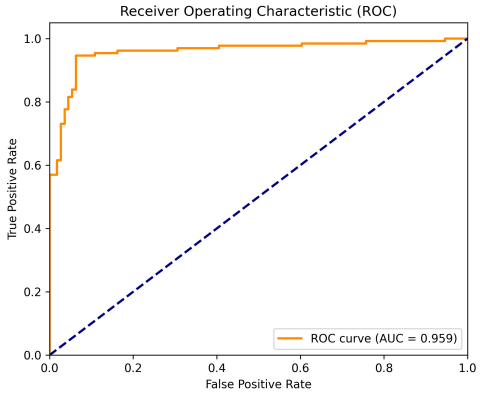


Figure 7: P-gp ROC curve from the repository's XGBoost evaluation. This is distinct from the graph-model out-of-fold results in Table 3.

5 Results Across Models

5.1 Association between measured endpoint labels

Among the 216 dual-labeled molecules, 45 were labeled $BBB^+/P\text{-}gp^+$, 95 $BBB^+/P\text{-}gp^-$, 19 $BBB^-/P\text{-}gp^+$ and 57 $BBB^-/P\text{-}gp^-$. The observed proportion carrying a positive P-gp label was 32.1% among BBB-positive compounds (45/140; exact binomial 95% CI 24.5–40.6%) and 25.0% among BBB-negative compounds (19/76). The table is imbalanced in a way consistent with only weak association: $\chi^2 = 0.887$, $p = 0.346$, Fisher odds ratio 1.42 (95% CI 0.76–2.67), and Cramér's $V = 0.064$. The interval remains wide enough to include moderate associations; the data support the claim that BBB status is not a reliable substitute for the P-gp label, not a claim of proven independence.

5.2 Single-split graph comparisons

The graph experiments evaluated random and scaffold-separated partitions with the same architecture family. On the random split, BBB ROC-AUC was 0.9388 for the single-task model and 0.9404 for multi-task learning, while P-gp ROC-AUC was 0.9390 and 0.9301, respectively. On the scaffold split, BBB ROC-AUC was 0.7583 versus 0.7467, and P-gp ROC-AUC was 0.8227 versus 0.8171. Thus, hard sharing produced no consistent gain across tasks or partition types. The scaffold GNN test sets contained 1,036 BBB labels and 333 P-gp labels, larger than the 48 dual-labeled records in that test set; joint quadrant metrics for this scaffold split were not computed.

Table 4: Single-split graph-model ROC-AUC comparisons from the encoded-feature runs.

Task	Split	Single-task	Multi-task	Difference
BBB	Random	0.9388	0.9404	+0.0016
P-gp	Random	0.9390	0.9301	−0.0089
BBB	Scaffold	0.7583	0.7467	−0.0116
P-gp	Scaffold	0.8227	0.8171	−0.0056

The random-to-scaffold decrease in the single-task GNN was 0.1805 ROC-AUC points for BBB and 0.1163 for P-gp. This indicates that both axes become harder under the scaffold

protocol, with a larger observed loss for BBB. These split-specific results are separate from the pooled random OOF metrics and from the descriptor-model scaffold results below; they do not establish that P-gp generalizes universally better across chemical space.

5.3 Descriptor-model validation

Table 5 summarizes the repository's descriptor-based XGBoost evaluations. The BBB model retains a high ROC-AUC under the saved scaffold split, but its specificity falls to 0.5795 and balanced accuracy to 0.7605. This illustrates why ranking metrics and the chosen classification threshold answer different questions. The P-gp scaffold result is also dataset- and protocol-specific and should not be generalized beyond the evaluated held-out set. Since the scaffold evaluation used a fresh untuned model, the random/scaffold contrast is descriptive rather than a strictly matched comparison.

Table 5: Selected metrics for the repository's XGBoost validation splits. Each cell is a point estimate; ROC-AUC 95% bootstrap CIs are shown in brackets.

Endpoint	Split	ROC-AUC [95% CI]	PR-AUC	Bal. Acc.	Specificity	MCC
BBB	Random	0.9522 [0.9410–0.9629]	0.9724	0.8775	0.8478	0.7535
BBB	Scaffold	0.9048 [0.8835–0.9240]	0.9668	0.7605	0.5795	0.5718
P-gp	Random	0.9587 [0.9330–0.9806]	0.9685	0.9287	0.9189	0.8580
P-gp	Scaffold	0.9631 [0.9359–0.9846]	0.9801	0.8978	0.9800	0.7843

5.4 Repeated-seed analysis and task weighting

To assess run-to-run variability, five random seeds were paired across single- and multi-task training. Mean ROC-AUC differences favored the single-task model in both endpoints, but confidence intervals on the mean differences overlapped zero: BBB $\Delta = -0.0094$ (95% CI -0.0234 to 0.0045; paired $p = 0.133$) and P-gp $\Delta = -0.0027$ (95% CI -0.0348 to 0.0294; paired $p = 0.826$). With only five paired seeds, these comparisons are underpowered to resolve small AUC changes. The same runs showed a consistent threshold-metric disadvantage for shared learning on BBB: paired differences in MCC, F1 and balanced accuracy were negative, with paired tests $p = 0.0059$, 0.0081 and 0.0097, respectively. Those threshold results are exploratory and should be interpreted alongside the small seed count and the validation-selected thresholds.

Table 6: Five-seed paired random-split ROC-AUC contrasts (multi-task minus single-task).

Endpoint	Mean difference	95% CI	Paired p	MDE	$P(\text{MT} > \text{ST})$
BBB	−0.00944	[−0.02337, 0.00449]	0.1331	0.01865	0.20
P-gp	−0.00272	[−0.03479, 0.02936]	0.8256	0.04295	0.54

The minimum detectable effect (MDE) is the estimated difference resolvable with 80% power under the recorded paired-seed variability and the stated testing assumptions. It is a design diagnostic, not a post-hoc guarantee. The estimated MDE exceeded the absolute observed AUC difference for both tasks, so a failure to reject a difference cannot establish equivalence. Conversely, the repeated-seed threshold metrics suggest the BBB decision boundary can be more sensitive to parameter sharing than its rank ordering.

The P-gp task was upweighted to test whether its smaller sample size explained the absence of transfer. Increasing its loss weight to 3 or 5 did not rescue P-gp ROC-AUC and reduced BBB performance. A zero-weight sanity run left the P-gp head near chance while retaining BBB performance, supporting the claim that task weighting was active as configured.

Table 7: P-gp loss-weight sensitivity on the random GNN split. The zero-weight run is an implementation check, not a valid P-gp predictor.

P-gp task weight	BBB ROC-AUC	P-gp ROC-AUC	Interpretation
0	0.9411	0.5626	P-gp head receives no task loss
1	0.9404	0.9301	Equal-weight multi-task baseline
3	0.9310	0.9295	No P-gp recovery; BBB declines
5	0.9086	0.9299	Greater BBB degradation

5.5 Feature ablation

The descriptor ablation tested whether added feature blocks improved discrimination in the XGBoost experiments. For BBB, combining Morgan fingerprints with six 2D descriptors gave ROC-AUC 0.9467, exceeding either the 2D-only model (0.9174) or Morgan-only model (0.9393). For P-gp, Morgan plus 2D features reached 0.9509; the full 2D+3D+fingerprint representation reached 0.9481. Three-dimensional shape descriptors alone were informative (0.8425) but did not improve the full P-gp result when appended to the fingerprint-plus-2D baseline. Differences are modest relative to their reported fold variation; these values do not prove that 3D shape has no value for any compound or use case.

Table 8: Feature-ablation ROC-AUC (mean ± standard deviation) from the repository experiments.

Endpoint	Feature configuration	Features	ROC-AUC
BBB	2D descriptors	6	0.9174 ± 0.0081
BBB	Morgan fingerprints	2,048	0.9393 ± 0.0054
BBB	Morgan + 2D	2,054	0.9467 ± 0.0035
P-gp	2D descriptors	6	0.9185 ± 0.0107
P-gp	3D shape descriptors	6	0.8425 ± 0.0243
P-gp	Morgan + 2D	2,054	0.9509 ± 0.0070
P-gp	Morgan + 2D + 3D	2,060	0.9481 ± 0.0076

Table 9: Joint decision results on 216 dual-labeled molecules using random-split OOF predictions. Bootstrap intervals use 1,000 resamples.

Decision strategy	Exact match (95% CI)	Q2 recall (95% CI)	Q2 leakage (95% CI)
BBB-only; P-gp assigned negative	55.6% (47.7–62.0)	0.0% (0.0–0.0)	93.3% (86.5–100)
Composed independent GNNs	57.9% (50.9–64.8)	44.4% (29.5–60.5)	48.9% (34.2–63.9)
Shared-encoder multi-task GNN	56.0% (49.5–62.5)	51.1% (36.6–66.7)	42.2% (26.7–56.8)

5.6 Four-quadrant error structure

Overall exact match obscures which decision is wrong. Table 10 therefore reports the full four-class confusion matrices, with rows representing observed quadrants and columns representing predictions. The BBB-only baseline cannot predict Q2 or Q4 by construction. It assigns every molecule a negative P-gp label, so its 42 Q2-to-Q1 errors are the direct consequence of substituting permeability for efflux. The two-axis models reduce that failure but still misclassify approximately half of Q2: 22 cases for composed independent models and 19 for the shared model. The multi-task model's slightly lower leakage comes with a lower exact match than the composed single-task predictions; the results do not establish an overall superiority of either two-axis architecture.

Table 10: Out-of-fold four-quadrant confusion matrices on the shared set. Rows are observed quadrants; columns are predicted quadrants.

Strategy	Observed	Pred Q1	Pred Q2	Pred Q3	Pred Q4
BBB-only	Q1	84	0	11	0
	Q2	42	0	3	0
	Q3	21	0	36	0
	Q4	5	0	14	0
Composed	Q1	66	18	8	3
	Q2	22	20	3	0
	Q3	16	5	33	3
	Q4	2	3	8	6
Multi-task	Q1	64	18	11	2
	Q2	19	23	2	1
	Q3	19	2	29	7
	Q4	4	5	5	5

The BBB score was further stratified by the observed P-gp class. For the composed independent models, BBB ROC-AUC was 0.7990 among P-gp-negative molecules ($n = 152$) and 0.9368 among P-gp-positive molecules ($n = 64$), a difference of -0.1379 (95% bootstrap CI -0.2406 to -0.0410 when computed as non-substrate minus substrate). For the multi-task model the corresponding values were 0.7904 and 0.8760, with a difference of -0.0856 (95% CI -0.1992 to 0.0247). This exploratory analysis runs counter to the simple expectation that efflux-positive molecules must be harder for a BBB classifier to rank. Label-stratified performance is influenced by chemical composition and the small number of shared compounds; it does not measure net exposure or establish a causal effect of efflux on the permeability label.

In the dual-labeled set, the BBB-positive subset carrying a positive P-gp label had higher mean molecular weight (370.9 versus 281.4 Da), higher LogP (3.8 versus 2.2), lower TPSA (47.8 versus 60.3 $Å^2$), fewer hydrogen-bond donors (0.8 versus 1.4) and more rotatable bonds (5.2 versus 3.9) than the P-gp-negative subset. These descriptive differences help interpret the stratified score result but are not adjusted causal estimates. They also illustrate why a model can correctly separate a permeability label and still omit the active-efflux decision.

5.7 Decision regret under varying progression costs

Equation 2 makes the trade-off between false progression and missed Q1 opportunities explicit. At $\lambda = 1$, regret was 0.3657 for BBB-only, 0.3194 for the composed model and 0.3380 for multi-task learning. As λ increases, the cost of advancing a false candidate rises, and the two-axis strategies become preferable. The crossover with BBB-only occurs near $\lambda = 0.643$ for the

composed strategy and 0.769 for the multi-task strategy in this sample. These crossovers are conditional on the stated binary loss and observed error counts; they must not be interpreted as clinically estimated cost ratios.

Table 11: Normalized decision regret at selected false-progression cost ratios. Lower values are better only under the stated cost function.

λ	BBB-only	Composed	Multi-task
0.25	0.1296	0.1806	0.1921
0.50	0.2083	0.2269	0.2407
1.00	0.3657	0.3194	0.3380
2.00	0.6806	0.5046	0.5324
4.00	1.3102	0.8750	0.9213
8.00	2.5694	1.6157	1.6991
16.00	5.0880	3.0972	3.2546

Repository TreeSHAP summaries ranked TPSA highest among the listed shared descriptors for the BBB model (mean absolute SHAP 0.9875) and MolLogP highest for the P-gp model (1.4074); MolWt was also attributed in both (BBB 0.6955, P-gp 0.6292). These are model-specific feature attributions on the reported scale, not causal effects or evidence of transporter mechanism.

The repository also screened 2,405 approved compounds and assigned 273 non-CNS-labeled drugs to the predicted BBB$^+$/P-gp-negative quadrant. These are in-silico prioritization results, not validated repurposing candidates; substantial overlap with benchmark data means the screen is not an external test. The full predicted quadrant distribution and its overlap with the training benchmarks are reported below.

6 Visual Results

The following panels reproduce the project's current figure set. They show model explanations, feature ablation, classification diagnostics, the approved-drug landscape and the structural docking analysis. Plots are descriptive: explanation scores are not causal effects, and docking scores are not experimental binding affinities.

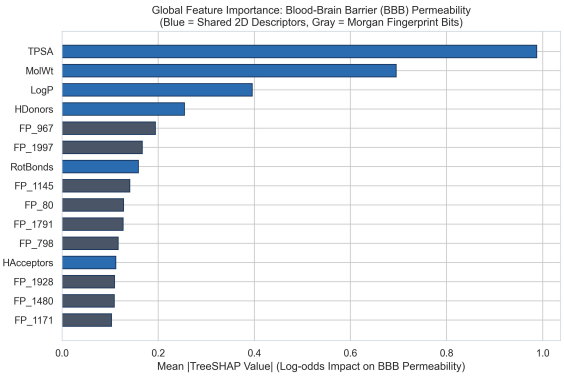


Figure 8: Global feature attributions for the BBB model.

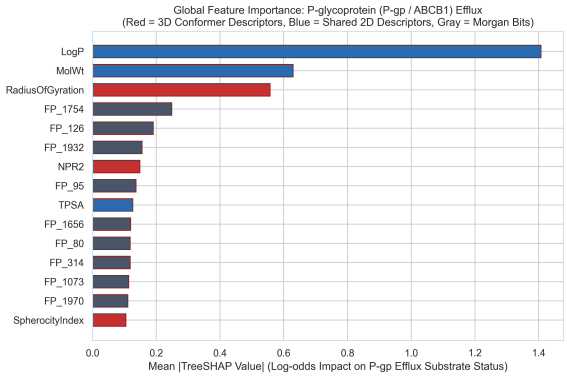


Figure 9: Global feature attributions for the P-gp model.

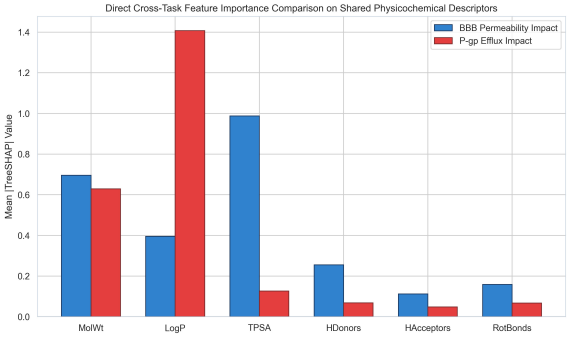


Figure 10: Comparison of shared molecular descriptors across tasks.

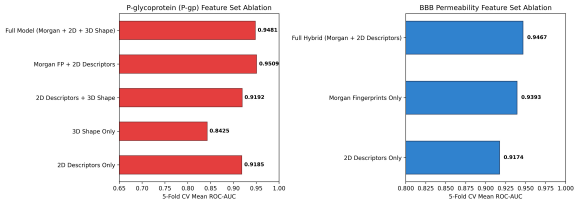


Figure 11: Feature-set ablation results.

6.1 Screening and structural-analysis figures

Predicted quadrants and disease-area distributions from the 2,405-compound repository screen are shown below. The docking figures summarize the repository's calculations against human P-gp structure 6QEX [7, 23, 24]. They are computational comparisons and do not establish transporter activity in a biological assay.

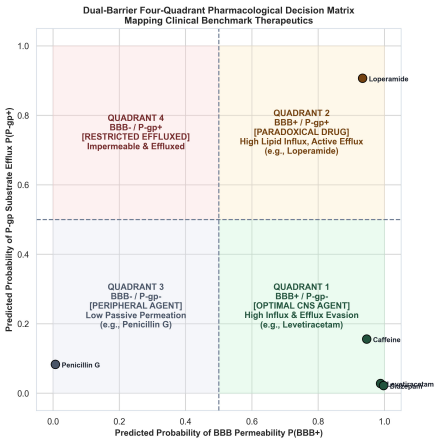


Figure 12: Joint BBB/P-gp quadrant distribution.

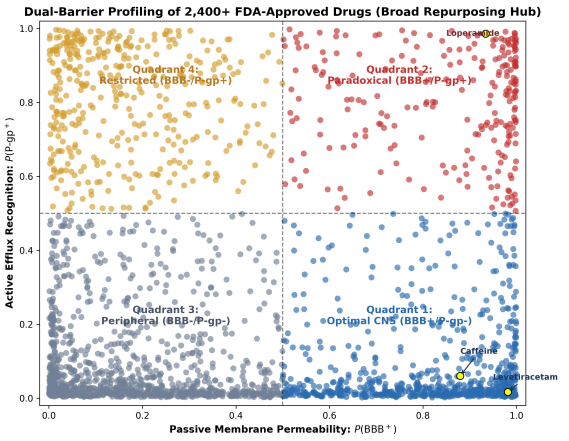


Figure 13: Predicted dual-barrier landscape of approved compounds.

Figure asset unavailable: `repurposing_disease_area_distribution.png`

Figure 14: Disease-area distribution in the repository screening results.

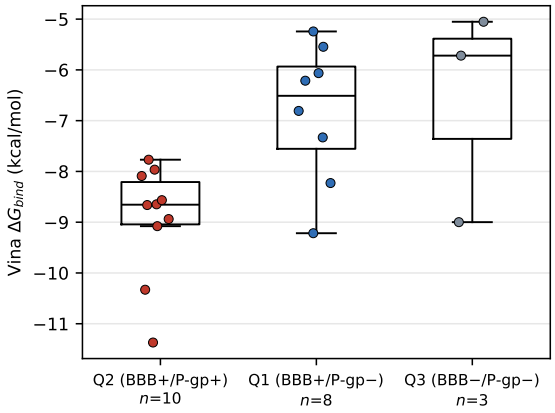


Figure 15: Docking-score distribution for the selected compounds.

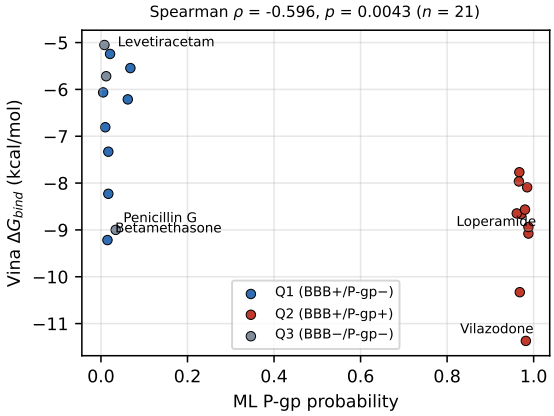


Figure 16: Relationship between model output and docking score.

6.2 Screening output and dataset overlap

The virtual screen assigned 710 compounds to Q1, 240 to Q2, 1,064 to Q3 and 391 to Q4. Of these, 273 compounds annotated as non-CNS approved drugs were included in Q1 and retained as hypotheses for further evaluation. The result is a prioritization list rather than a repurposing claim: benchmark training overlap, domain distance and known pharmacology must be checked compound by compound.

The repository audit reports that 1,330 screened molecules occur in B3DB and 354 occur in the P-gp model's training data. Among the Q1 non-CNS leads, 160 overlap the B3DB benchmark; among the 105 marketed neuropsychiatric compounds assigned to Q2, 27 occur in the P-gp training set. These overlaps limit the screen's evidential independence. The classifications can still be operationally useful for inspecting pipeline behavior, but a prospective or truly external candidate list must exclude known benchmark compounds and be evaluated with experimentally

Table 12: Predicted quadrants in the 2,405-compound approved-drug screen.

Quadrant	Predicted interpretation	Count	Fraction
Q1	$BBB^+/P\text{-}gp^-$	710	29.5%
Q2	$BBB^+/P\text{-}gp^+$	240	10.0%
Q3	$BBB^-/P\text{-}gp^-$	1,064	44.2%
Q4	$BBB^-/P\text{-}gp^+$	391	16.3%
Total		2,405	100.0%

independent measurements.

Table 13: Known overlap between the approved-drug screen and model benchmark sets, as recorded in the repository audit.

Screen subset	Benchmark overlap	Total subset
All approved compounds in B3DB	1,330	2,405
All approved compounds in P-gp training data	354	2,405
Predicted Q1 non-CNS leads in B3DB	160	273
Predicted Q2 neuropsychiatric compounds in P-gp training data	27	105

6.3 Feature attribution and interpretation

TreeSHAP values explain the trained tree ensemble's predictions in its output-margin space [25]. The global BBB summary ranks TPSA among the leading descriptors, whereas the P-gp model places LogP prominently. In the repository's quantitative comparison of shared properties, mean absolute SHAP was 0.9875 for BBB TPSA and 1.4074 for P-gp MolLogP; the corresponding values for MolWt were 0.6955 and 0.6292. These summary magnitudes are model- and scale-dependent. They do not constitute causal effects, identify a unique transporter-recognition motif, or say that lowering TPSA or changing LogP will improve a compound without affecting other properties.

The feature-attribution analysis also has the usual limitations of correlated descriptors and fingerprint features. When correlated inputs share predictive information, attribution can be distributed among them in a model-dependent way. A global mean absolute attribution removes the sign and averages across molecules, so it cannot reveal whether a feature raises or lowers a prediction in a particular chemical series. Local explanation and a substructure mapping can support medicinal-chemistry review, but any proposed edit remains a computational hypothesis requiring synthesis and measurement.

6.4 Structural docking as exploratory analysis

The repository docked a small set of selected molecules against the human P-gp cryo-electron microscopy structure 6QEX [7, 23, 24, 26]. Docking scores and residue-contact counts summarize poses under a prepared, largely fixed receptor model. They are not experimentally measured binding free energies, transport rates or proof that a molecule is a substrate; substrate and inhibitor binding also cannot be separated solely by a favorable score in one receptor conformation.

The raw docking CSV contains 22 rows; the accompanying complete-case analysis reports

$n = 21$. In that audit, heavy-atom count was strongly associated with more favorable docking score (Spearman $\rho = -0.817$, $p < 10^{-5}$) and with model P-gp probability ($\rho = 0.757$, $p < 10^{-4}$). After controlling for heavy-atom count using the reported partial-rank analysis, the association between predicted P-gp probability and docking energy was near zero ($\rho = 0.060$, $p = 0.798$). Accordingly, the apparent raw relationship cannot be presented as independent structural confirmation of the classifier.

The chosen receptor conformation, protonation and charge assignment, ligand conformer generation, search box and scoring function all influence the resulting score. The analysis also has small sample size, possible benchmark overlap, and a size-composition imbalance between predicted groups. A stronger structural study would use an independently selected and size-matched compound set, multiple receptor conformations, replicate docking settings and experimental transporter assays. Docking is retained here as a pipeline demonstration and a cautionary example of confounding, not as validation.

7 Discussion

7.1 Two distinct biological axes should remain distinct in the model

The dual-label subset contains clear counterexamples to the idea that BBB-positive status is interchangeable with efflux-negative status: 45 of 140 BBB-positive molecules (32.1%) are also annotated P-gp-positive. The weak association statistics reinforce that the two benchmark labels should be measured and reported separately. The label count does not establish independence in the population, and the limited overlap leaves wide uncertainty; nevertheless, the practical finding is robustly framed as a prevalence in this sample rather than an overinterpreted significance test.

The user-confirmed interpretation of P-gp-positive as an efflux substrate is biologically coherent with the four-quadrant motivation. However, the exact benchmark lineage remains a central reproducibility issue: the referenced Broccatelli publication describes inhibition prediction, whereas the project's task is presented as substrate status. This paper reports the project-defined label semantics and the computed results, but it cannot independently certify that the raw TDC target was mapped correctly. A conference submission should cite the exact TDC task/version and preserve the upstream label mapping or identify a substrate assay source.

7.2 Per-task discrimination does not equal joint decision accuracy

The graph models attain high pooled OOF ROC-AUC on each individual endpoint, yet the two-axis exact-match rate on the 216 shared molecules is only 55.6–57.9%. These are not contradictory measures: ROC-AUC evaluates ranking on one axis over thresholds, whereas exact match requires two binary calls on the same molecule to be simultaneously correct. A favorable BBB score therefore cannot answer whether the compound is effluxed.

The BBB-only baseline makes the implication explicit. By assigning P-gp-negative to every molecule, it has zero recall for Q2 and leaks 42 of 45 Q2 molecules into Q1. The two-model and shared-encoder strategies reduce leakage to 22 and 19 molecules respectively, but their confidence intervals overlap and their overall exact-match estimates differ by only a few percentage points. The central practical implication is to report the efflux axis and the joint error profile alongside permeability metrics, rather than to claim that a second classifier solves the decision problem.

Quadrant calls should also be presented with their component probabilities and, where possible, uncertainty or applicability-domain flags. The binary cut at the default prediction threshold can obscure borderline molecules, and the current cross-task system has not been calibrated jointly against an in vivo exposure endpoint. A Q1 call is thus a screening category, not a clinical recommendation.

7.3 Interpretation of the multi-task result

The transfer hypothesis was plausible because the BBB-labeled corpus is substantially larger than the P-gp-labeled corpus and both tasks are defined over molecules. Empirically, pooled OOF ROC-AUC changed by -0.0016 for BBB and $+0.0045$ for P-gp under multi-task learning. Single-split results show the absence of a consistent advantage across random and scaffold settings, while five-seed comparisons show that the AUC contrasts remain imprecise. The defensible conclusion is that these experiments did not demonstrate a reliable ROC-AUC benefit from hard parameter sharing.

The lower BBB MCC, F1 and balanced accuracy across five paired seeds raise a separate, threshold-sensitive concern: shared learning may make the BBB decision boundary worse even when AUC is similar. Five seeds are not enough to establish the magnitude or universality of that effect, but this evidence cautions against using a single ranking metric to conclude that the models are interchangeable. More task weighting did not restore P-gp performance and coincided with lower BBB AUC. The zero-weight control produced a near-chance P-gp score while retaining BBB performance, reducing concern that the task-weight implementation was inert.

The near-zero label association is a possible explanation for limited transfer, but it does not prove a mechanism of negative transfer. Label correlation is only one factor in multi-task learning; shared low-level chemistry, task noise, differences in assay provenance, representation capacity, optimization, and sampling can also determine whether shared parameters help. A future design could test soft sharing, task-specific adapters, uncertainty-weighted losses, or a larger curated dual-label set, but those alternatives were not evaluated here.

7.4 Generalization and threshold behavior

The scaffold experiments emphasize the importance of evaluation protocol. For the single-task graph models, scaffold-split ROC-AUC was lower than random-split ROC-AUC for both endpoints, more strongly for BBB. The XGBoost BBB scaffold ROC-AUC is still relatively high, but the thresholded specificity falls to 0.5795. That combination demonstrates how a model can preserve ranking while assigning too many negatives to the positive class at a fixed threshold. The random and scaffold comparisons are not fully matched because the descriptor validation trains a fresh model and the graph test partitions are independent; they should be reported as separate pieces of evidence.

The single-split scaffold P-gp test set includes 333 labeled structures and the BBB test set 1,036; the dual-labeled subset is only 48. The joint OOF analysis, by contrast, uses a random five-fold protocol. Thus, the paper does not claim that its quadrant leakage figures persist under scaffold shift. Future work must compute the same joint decision metrics on scaffold-held-out dual-labeled compounds and preferably use an external chemical series.

7.5 Feature representation and explanations

The feature ablation supports a practical, not mechanistic, conclusion: Morgan fingerprints plus simple 2D descriptors form a strong baseline for both tasks, and the additional 3D shape descriptors did not improve the reported P-gp ROC-AUC. The fact that 3D-only descriptors achieved 0.8425 suggests that shape encodes some signal in this benchmark, but the full feature-set comparison does not justify claiming that 3D modeling improved prediction. Nor does the comparison rule out value from conformer ensembles, transporter-specific structural features or improved assay labels.

The SHAP summary suggests different predictive associations across the two tree models, with TPSA prominent in the BBB attribution and LogP prominent for P-gp. This pattern is broadly compatible with the conceptual distinction between desolvation/permeability and transporter recognition, but the model explanation is not a mechanistic experiment. Correlated descriptor effects, training-set composition and label provenance limit biological interpretation. The SHAP figures are best used to audit model behavior and formulate hypotheses for subsequent testing.

7.6 Virtual screening and structural analyses

The approved-drug screen demonstrates that the trained classifiers can be applied at scale and organize compounds into four interpretable categories. Its value is hypothesis generation; the large benchmark overlap means it is not an independent validation of the candidate list. A stronger repurposing analysis would conduct a leave-known-training-compounds-out screen, state the applicability-domain status, compare against established CNS annotations and evaluate a blinded set experimentally.

Similarly, exploratory docking should not be used as a second line of evidence for the classifier when heavy-atom count explains a substantial part of the raw score association. The size-adjusted correlation reported by the repository is close to zero. The use of one prepared structure and approximate docking score adds structural context, but does not discriminate substrate from inhibitor or establish active transport. This more restrained interpretation is important for conference review because the raw group-separation numbers can appear persuasive while being confounded.

8 Limitations and Future Work

8.1 Endpoint mapping and assay harmonization

The most important limitation is the provenance of the P-gp label. Although the project authors define P-gp-positive as efflux/substrate-positive, the cited Broccatelli paper's title describes inhibitor prediction. The manuscript therefore distinguishes project-specific label semantics from independently verified assay semantics. The precise raw file, TDC release, target name, positive-class conversion and assay definition must accompany any public release of the 216-molecule overlap set. A carefully curated substrate-specific benchmark should be used to validate the key result before the physiological interpretation is generalized.

The BBB and P-gp datasets were curated separately and may differ in assay systems, protocols, compound sourcing and class thresholds. A binary model cannot recover quantitative

permeability or transport kinetics from labels alone. The small dual-labeled set is vulnerable to sampling variation and represents only a narrow intersection of the two source collections. Larger, harmonized data with experimental metadata and matched compounds are required.

8.2 Evaluation design and statistical power

The OOF analysis covers all molecules under a random split, but the resulting folds may share close analogs across train and test. The scaffold test reduces this risk for per-task model evaluation, yet the paper lacks scaffold OOF quadrant metrics and a prospective external test. Five repeated seeds provide only limited power to resolve small AUC differences; the MDE estimates illustrate this limitation. Bootstrap intervals on the same test compounds quantify sampling uncertainty but do not replace independent validation, correct for all model-selection choices, or account for dataset shift.

The association analysis is likewise underpowered to establish a small relationship or equivalence. The odds-ratio interval from 0.76 to 2.67 includes both near-null and moderately elevated associations. Accordingly, we do not use the phrase "statistically independent." The observed proportions and effect sizes should be re-estimated on a larger, independently curated overlap.

8.3 Biological scope and clinical translation

BBB permeability and P-gp substrate status are components of brain exposure, not complete determinants of pharmacological efficacy. Transport by BCRP and other ABC transporters, passive influx, plasma and brain binding, metabolism, molecular ionization, dosing, species differences and target potency all contribute. The project does not measure unbound brain-to-plasma partition coefficient ($K_{\mathrm{p,uu}}$), transporter kinetics, exposure-response or therapeutic outcome. A Q1 prediction cannot establish that a compound reaches an effective concentration or is safe for a CNS indication.

No external in vitro transport experiment was performed in this study, and docking is not a substitute for such an experiment. The screened compounds may be outside model applicability domains or may have adverse properties unrelated to BBB penetration. Candidate-specific claims should be deferred until those issues are evaluated.

8.4 Prioritized next steps

The next analysis should verify the P-gp source and class mapping; produce a versioned, deduplicated release of the 216 shared molecules; and compute scaffold-held-out quadrant metrics with fold integrity checks. A second priority is an external test using an explicitly substrate-labeled P-gp dataset and measured BBB endpoints. The model comparison should expand paired seeds and include calibration, threshold selection confined to validation folds, and pre-specified joint costs. Finally, candidate compounds should be tested in transporter assays and, where feasible, against unbound brain exposure; docking should be repeated on size-matched compounds and multiple receptor conformations only if it informs a clearly stated structural hypothesis.

9 Conclusion

This paper evaluates a two-axis CNS screening problem using project-defined BBB and P-gp labels, compares separate and shared-encoder graph models, and measures the compound-level

decisions obscured by single-axis scores. The two labels show weak observed association in the 216-molecule intersection, and hard parameter sharing does not show a reliable pooled OOF ROC-AUC advantage. Per-axis discrimination nevertheless coexists with modest exact quadrant accuracy and substantial Q2-to-Q1 leakage under the BBB-only baseline. Reporting the P-gp axis alongside BBB permeability therefore makes a materially different decision visible.

The conclusion is deliberately bounded: the data do not establish label independence, a clinical exposure prediction, or a superiority of one two-axis model. The project-defined P-gp semantics are accepted here following the authors' clarification, but the source mapping remains a reproducibility caveat. Larger dual-labeled benchmarks, external scaffold validation and experimental transporter/brain-exposure measurements are necessary before repurposing or clinical claims can be made.

Data and code availability

Project code, benchmark-processing scripts, model artifacts, figures and result summaries are available in the public repository [27]. The repository includes the OOF prediction files and integrity reports needed to reproduce the joint-decision calculations. Raw upstream data remain subject to their original source licenses and terms. The manuscript's numerical claims should be read alongside the specific saved artifact and split protocol named in the corresponding table caption.

A Metric definitions and quadrant accounting

Let y_i and $\hat{y}_i$ denote the observed and predicted binary outcomes for a task. Sensitivity is $\mathrm{TP}/(\mathrm{TP}+\mathrm{FN})$, specificity is $\mathrm{TN}/(\mathrm{TN}+\mathrm{FP})$, balanced accuracy is the average of sensitivity and specificity, and MCC summarizes all four entries of the binary confusion matrix. ROC-AUC estimates the probability that a randomly selected positive receives a higher score than a randomly selected negative. PR-AUC summarizes the precision–recall curve and depends on the prevalence of positives in the evaluated sample.

For the joint evaluation, quadrant exact match is

$$\mathrm{QEM} = \frac{1}{N}\sum_{i=1}^{N} \mathbb{1}[\hat{q}_i = q_i]\,,$$

where q_i is the observed combination of the two binary labels. Q2 recall is TP_{Q2}/N_{Q2}. Q2 leakage is

$$\mathrm{Leak}_{Q2\to Q1} = \frac{\#\{i : q_i = Q2,\ \hat{q}_i = Q1\}}{N_{Q2}}.$$

Leakage is a subset of Q2 false negatives, but not all missed Q2 predictions are Q1: some are assigned to Q3 or Q4. This distinction explains why paradox recall and leakage need not sum to one.

B Reproducibility and experiment provenance

The principal graph experiments used the encoded 66/14 atom/bond featurization, 256 hidden channels, three GINE layers and seed 42 for the primary fixed-split and five-fold runs. The OOF

Table 14: Operational definitions of the four quadrants used throughout the analysis.

Quadrant	BBB label	P-gp label	Screening interpretation
Q1	Positive	Negative	Permeable; no labeled efflux liability
Q2	Positive	Positive	Permeable; labeled efflux substrate
Q3	Negative	Negative	Permeability-limited under these labels
Q4	Negative	Positive	Both permeability-negative and efflux-positive

report records fold assignments, best epochs, training configuration, class weights, hardware and data cardinalities. Five-seed variance analysis used paired seeds 42–46 and a random split. The reported task-weight sensitivity runs altered only the P-gp loss multiplier in the corresponding training configuration.

The repository's five-fold OOF integrity report records 8,801 molecules scored exactly once, all 216 dual-labeled compounds scored on both axes, and zero train/test prediction leaks. For the single-task runs, all 7,805 BBB labels and 1,212 P-gp labels were covered. These checks address bookkeeping integrity; they do not guarantee absence of close analogs across random folds or remove overlap between the approved-drug screen and model training data.

Table 15: Primary result artifacts and the claim each supports.

Artifact	Manuscript use
`gnn_single_bbb_kfold5_encoded_report.json`	Single-task BBB pooled OOF metrics
`gnn_single_pgp_kfold5_encoded_report.json`	Single-task P-gp pooled OOF metrics
`gnn_multitask_kfold5_encoded_report.json`	Shared-encoder pooled OOF metrics and integrity checks
`gnn_decision_metrics_report.json`	Quadrant, leakage, regret and stratified AUC summaries
`gnn_experiment_comparison.csv`	Fixed random/scaffold graph comparisons
`gnn_seedvar_random_report.json`	Paired five-seed uncertainty and MDE summaries
`validation_metrics_summary.csv`	Descriptor-model random/scaffold metrics
`ablation_study_results.csv`	Descriptor/fingerprint/3D feature ablations
`shared_properties_quantitative_comparison.csv`	Shared-descriptor SHAP summaries
`cryoem_pgp_docking_validation.csv`; `extra_analysis.json`	Exploratory docking and size-adjusted association

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